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REPORT DOCUMENTATION PAGE

READ INSTRUCTIONS BEFORE COMPLETING FORM

1. REPORT NUMBER: TOP 2-2-508	2. GOVT ACCESSION NO.: A121978
3. RECIPIENT'S CATALOG NUMBER:	
4. TITLE *(and Subtitle)*: US ARMY TEST AND EVALUATION COMMAND TEST OPERATIONS PROCEDURE "AUTOMOTIVE SAFETY AND HEALTH HAZARD EVALUATION"	5. TYPE OF REPORT & PERIOD COVERED: Final
	6. PERFORMING ORG. REPORT NUMBER:
7. AUTHOR(s):	8. CONTRACT OR GRANT NUMBER(s):
9. PERFORMING ORGANIZATION NAME AND ADDRESS: US ARMY ABERDEEN PROVING GROUND (STEAP-MT-M) ABERDEEN PROVING GROUND, MARYLAND 21005	10. PROGRAM ELEMENT, PROJECT, TASK AREA & WORK UNIT NUMBERS: DARCOM-R 310-6
11. CONTROLLING OFFICE NAME AND ADDRESS: US ARMY TEST AND EVALUATION COMMAND (DRSTE-AD-M) ABERDEEN PROVING GROUND, MARYLAND 21005	12. REPORT DATE: 24 November 1982
	13. NUMBER OF PAGES: 21
14. MONITORING AGENCY NAME & ADDRESS *(if different from Controlling Office)*:	15. SECURITY CLASS. *(of this report)*: Unclassified
	15a. DECLASSIFICATION/DOWNGRADING SCHEDULE:

16. DISTRIBUTION STATEMENT *(of this Report)*

Approved for public release; distribution unlimited.

17. DISTRIBUTION STATEMENT *(of the abstract entered in Block 20, if different from Report)*

18. SUPPLEMENTARY NOTES

19. KEY WORDS *(Continue on reverse side if necessary and identify by block number)*

Brakes	Health	Standards	Test Methods
Classification	Noise	Steering	Vehicles
Hazards	Safety	Storage	

20. ABSTRACT *(Continue on reverse side if necessary and identify by block number)*

Procedures are described to identify and evaluate real and potential safety and health hazards that exist in military tracked and wheeled vehicles. Referenced test procedures are taken in part from Federal Motor Vehicle Safety Standards and several TECOM Test Operation Procedures (TOPs). The safety tests include the following procedures to evaluate existing and potential hazards: static vehicular stability, braking, steering, human factors, sound level, toxic gas level, stowage, safety aspects of maintenance, weapon system safety, overhead guards, and FMVSS requirements.

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US ARMY TEST AND EVALUATION COMMAND
TEST OPERATIONS PROCEDURE

DRSTE-RP-702-102
*Test Operations Procedure 2-2-508 24 November 1982
AD No. A121978

AUTOMOTIVE SAFETY AND HEALTH HAZARD EVALUATION

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1. SCOPE. This TOP describes those procedures that are used to identify and evaluate real and potential safety and health hazards that exist in military tracked and wheeled vehicles. This TOP is applicable primarily to the automotive subsystem of vehicles, but it briefly mentions procedures for evaluating the weapon or other subsystems that may be associated with the vehicle.

2. FACILITIES AND INSTRUMENTATION. Facilities and instrumentation are described in references cited by this TOP.

3. REQUIRED TEST CONDITIONS.

3.1 Planning.

a. Review all instructional material issued with the test vehicles by the manufacturer, contractor, or government, as well as reports of previous similar tests on the same types of vehicles.

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b. Review the safety assessment report provided by the developer to ensure that the identified hazards have been adequately addressed in the test plan which shall, in turn, be written to ensure that subtests suitable for evaluating the identified hazards are included. In addition, review applicable SOPs to make sure they are suitable for the test design under consideration.

c. From Table 1, determine which safety requirements of MIL-STD-1180A(AT)[1]* apply to the vehicle being tested. Review the documents cited by MIL-STD-1180A(AT), identifying those for which compliance can be established by inspection or other planned tests, and those that will require separate additional testing. Data from other planned tests should be used to the maximum extent to reduce the scope of testing.

d. Compile a list of all safety- and health-related requirements from pertinent documents (standards, regulations, specifications, ROC, etc.). The checklists in Appendix A and TOP 10-2-508[2] can be used to compile such a list.

e. Based on the information assembled above, develop a comprehensive test program that includes the following essential features:

(1) Inspections and tests necessary to certify that the vehicle is safe for continued testing through a safety release prepared in accordance with AR 385-16.[3]

(2) Selected subtests to verify that the vehicle being tested satisfies design and functional requirements for safe field deployment. These tests are used to evaluate the vehicle against the demands of its proposed military mission, whereas the standards set forth in MIL-STD-1180A(AT) are taken from documents used to evaluate vehicles in a nonmilitary environment. Tests to be considered for an overall safety and health evaluation and their associated TOP/MTPs include:

Gradeability and Side Slope Performance - TOP 2-2-610[4]
Braking, Wheeled Vehicles - TOP 2-2-608[5]
Braking, Tracked Vehicles - TOP 2-2-627[6]
Steering - TOP 2-2-609[7]
Human factors evaluation - TOP 1-2-610[8]
Logistic support - TECOM Suppl 1 to DARCOM Reg 700-15[9]
Noise level - TOP 1-2-608[10]
Toxic hazards - TOP 2-2-614[11]
Field shock and vibration - TOP 2-2-808[12]
Fording - TOP 2-2-612[13]
Standard Obstacles - TOP 2-2-611[14]
Automotive Winches - TOP 2-2-712[15]
High and low temperature - TOP 2-2-816[16]
Mechanical, electrical, miscellaneous hazards - TOP 10-2-508
Night Performance - MTP 2-2-616[17]

(3) Systematic observations and analysis of the vehicle throughout all phases of development testing to identify and investigate any actual or potential hazards to personnel and equipment resulting from design deficiencies, personnel errors, environmental conditions, or system failures.

*NOTE: Footnote numbers correspond to reference numbers in Appendix C.

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Table 1. MIL-STD-1180A(AT) Applicability

Requirement No.	Subject Area	Vehicle Type[a]							
		I	II	III	IV	V	VI	VII	VIII
101	Control and displays	X	X		X	X	X	X	--
102	Transmission shift level sequence, starter interlock, and braking effect	X	X		X	X	X	X	--
103	Windshield defrosting/defogging systems	X	X		X	X	X	X	--
104	Windshield wipers and washers	X	X		X[b]	X[b]	X[b]	X	--
105	Hydraulic brake systems	X	X	X	X[c]	X[c]	X[c]	X	--
106	Brake hoses			All so equipped					
107	Reflective surfaces	X	X	X	X	X	X	X	X
108	Lamps, reflectors, and signaling devices	X	X	X	X	X	X	X	X
111	Rearview mirrors	X	X			X	X[c]	X	X
113	Hood latches	X	X					X	--
116	Hydraulic brake fluid		All with hydraulic brakes						
119	New pneumatic tires	X	X	X			X[c]	X	X
120	Tire selection and rims	X	X	X	X		X	X	X
121	Air brake systems	X[d]	X[de]	X[d]				X[d]	X
122	Motorcycle brake system								X
123	Motorcycle control and displays								X
124	Accelerator control systems	X	X		X	X	X	X	X
125	Warning devices	X	X				X[c]	X	--
201	Occupant protection in interior impact	X	X					X	--
203	Impact protection for driver from steering control system	X	X					X	--
204	Steering control rearward displacement	X	X					X	--
205	Glazing material			All so equipped					
206	Door locks & retention components	X	X					X	--
207	Seating systems	X	X					X	--
208	Occupant crash protection	X	X			X[c]	X[c]	X	--
209	Seatbelt assemblies			All so equipped					--
210	Seatbelt anchorages	X	X[c]			X[c]	X[c]	X	--
219	Windshield zone intrusion	X	X					X	--
301	Fuel system integrity	X	X		X	X	X[c]	X	--
302	Flammability of interior materials	X	X		X	X	X	X	--

[a] I Tactical, standard, wheeled vehicles
II High-mobility, tactical, wheeled vehicles
III Trailers
IV Combat vehicles
V Tactical, tracked vehicles
VI Construction & material-handling vehicles
VII Tactical-support, wheeled vehicles
VIII Motorcycles

[b] At discretion of developer

[c] When specified by procuring activity

[d] Unless waived by procuring activity

[e] For Type II vehicles driven at high speeds

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3.2 Automotive Subsystem.

a. Inspect the automotive subsystem in accordance with TOP 2-2-505[18].

b. Inspect the automotive subsystem for potential hazards using the checklist of Appendix A and the findings of the safety assessment report as a guide. Additional guidance may be obtained from TOP 10-2-508 to identify and evaluate electrical, mechanical and miscellaneous hazards.

3.3 Vehicle-Mounted Subsystem. Ensure that weapons and ammunition have already been standardized, or have received a safety evaluation as indicated by TOP 3-2-805[19] and TOP 4-2-504[20], respectively, and are considered safe.

3.4 Test Controls.

a. Test items in the configuration and condition in which they are expected to be deployed and operated by the field army. Modifications required to accommodate instrumentation should not affect the performance of the vehicle being tested.

b. Make sure the safety and health evaluation is planned, conducted, and reported by engineering personnel who are occupationally qualified in the specific commodity under test and coordinated with and supported by occupationally qualified safety engineers.

c. Make sure all precautions necessary to ensure highest feasible degree of safety of test participants and auxiliary equipment will be followed during all phases of testing.

d. Through briefings and the use of mockups or test items, orient test personnel regarding the test item and test activity safety precautions and procedures, normally as a part of the initial equipment training program.

4. TEST PROCEDURES. Generally, safety subtests are performed early in the test program so that hazards to test participants can be minimized, and a safety release recommendation can be prepared in accordance with TECOM Supplement 1 to AR 385-16[21]. The subtests, described in the following subparagraphs, should be integrated to the maximum extent with other planned tests to avoid duplication and reduce the duration of testing. Those subtests that are required for development tests are indicated below.

4.1 Federal Motor Vehicle Safety Standards (FMVSS) Actions (Configuration).

4.1.1 Method. Consult Tables 1 and 2 and MIL-STD-1180A(AT) for guidance in establishing criteria for the configuration of the vehicle being tested. The applicability of MIL-STD-1180A(AT) to each vehicle varies, and tests should be coordinated with the developing agency in advance. Compliance of the vehicle with many criteria can be ascertained by inspection or by a manufacturer's certification.

4.1.2 Data Required. These are indicated in the associated FMVSS.

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Table 2. Subtests for Conformance With FMVSS

Note: Inspection subtests apply to all FMVSS actions

FMVSS No.	Subject Area	Subtests		
		Configuration	Performance	Human Factors
101	Controls	-------	-------	X
102	Transmission	-------	X	-------
103	Windshield defrosting	-------	X	-------
104	Windshield wipers & washers	-------	X	-------
105	Hydraulic brakes	-------	X	-------
106	Brake hose & assemblies	-------	X	-------
107	Reflecting surfaces	-------	X	X
108	Lighting systems	-------	X	X
109	Pneumatic tires	X	-------	-------
110	Tire selection & rims	X	X	-------
111	Rearview mirrors	X	-------	-------
113	Hood latches	X	-------	-------
114	Theft protection	-------	-------	-------
115	Vehicle identification No.	-------	-------	-------
116	Hydraulic brake fluids	-------	X	-------
117	Retreaded tires	X	X	-------
121	Air brake systems	-------	X	-------
122	Motorcycle brake system	X	X	-------
123	Motorcycle controls and display	X	-------	-------
124	Accelerator control systems	-------	-------	-------
125	Warning devices	-------	-------	X
201	Crash protection	X	-------	-------
202	Head restraints	X	-------	-------
203	Steering control impact	-------	X	-------
204	Steering control displacement	X	-------	-------
205	Glazing materials	X	-------	-------
206	Door locks	X	-------	-------
207	Seats	X	-------	X
208	Seat-belt installation	X	-------	-------
209	Seat-belt assemblies	X	-------	-------
210	Seat-belt anchorage	X	-------	-------
212	Windshield mounting	X	-------	-------
214	Side-door strength	X	-------	-------
215	Exterior protection	X	-------	-------
219	Windshield zone intrusion	X	-------	-------
301	Fuel system	X	-------	-------
302	Flammability	X	-------	-------

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4.2 Static Stability (required).

4.2.1 Method. Test vehicles to determine their operating ability on longitudinal grades as steep as 60% and on side slopes as steep as 40%, in accordance with TOP 2-2-610. Conduct vehicle steering tests on side slopes, in accordance with TOP 2-2-609, including sine wave maneuvering of the vehicle perpendicular to the slope, exposing the vehicle to conditions potentially very near the critical stability angle for the vehicle.

By tests, computations, or both, and by using procedures described in TOP 2-2-800,[22] determine the combined center of gravity of the vehicle and load. From measurements such as these, make a preliminary estimate of the safe limits of stability while operating the vehicle.

Other factors to be considered include the following: overturning moments on curves, bounce from irregular terrain, nonuniform loading, tire compression on curves, and shifting of liquid cargo.

4.2.2 Data Required. Specific data requirements are described in the TOPs that are cited in this section. The data are assessed versus operational characteristics of the vehicle to establish the extent to which safe stability is maintained.

4.3 Braking (Required).

4.3.1 Method. This subtest is designed to determine the safety of the vehicle with respect to the brake system performance. Specific methods are provided in TOP 2-2-608 for wheeled vehicles and in TOP 2-2-627 for tracked vehicles.

4.3.2 Data Required. The data required for any of the specific braking subtests are listed in the cited TOPs. An assessment of how the braking performance compares with applicable criteria for that vehicle must be made. Any performance that does not meet or exceed criteria must be analyzed as to how it may adversely affect the safety of the vehicle.

4.4 Steering (Required).

4.4.1 Method. Select tests from TOP 2-2-609 to demonstrate steering control, particularly on slopes, at high speeds on paved surfaces, and when operating over rough terrain.

4.4.2 Data Required. Data include steering forces, steering response, turning diameters, degree of control during operating and maneuvering. Objective and subjective data are assessed to judge the safety of the vehicle steering performance.

4.5 Human Factors (Required).

4.5.1 Methods. Evaluate specific design features of the vehicle with respect to providing occupants a safe transport environment. The evaluation should include, but not be limited to the following: crew station anthropometrics; whole body vibration; visibility; human-machine interface; sound levels (see paragraph 4.6); and toxic gases (see paragraph 4.7). Specific tests described in TOP 1-2-610 can be used for evaluating the adequacy of safety provisions from the aspect of human

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factors engineering. Refer to Tables 1 and 2 to determine those FMVSS that are applicable to human factors considerations.

4.5.2 Data Required. Data include measurements of operating spaces; observations on the potential for system-induced human errors; vibration amplitude and frequency profile; sound-level measurements; and toxic-gas levels. Other data can be collected during the human factors subtest that may affect safety. All such data are compared against acceptable standards, and judgments are made in subjective areas as to the safety of the vehicle as designed.

4.6 Sound Level (Required).

4.6.1 Method. Measure vehicle noise at two-thirds of the rated engine speed in the highest gear for compliance with MIL-STD-1474B(MI)[23]. Also measure vehicle noise in all other gears at two-thirds of maximum road speed for that gear for informational purposes.

4.6.2 Data Required. Data to be recorded include sound pressure levels measured at normal crew positions and the average time of exposure. Measurements are made with the vehicle at two-thirds the off-highway payload.

4.7 Toxic Gases Measurement (Required).

4.7.1 Methods. Measure levels of carbon monoxide (CO), nitrogen dioxide (NO_2), and sulfur dioxide (SO_2) while the vehicle is operating, according to the procedure described in TOP 2-2-614.

4.7.2 Data Required. Description of test conditions, sources of contaminants, gas identity, and concentrations are recorded. The gas concentrations are compared as specified in TOP 2-2-614.

4.8 FMVSS[24] Actions (Performance).

4.8.1 Method. Consult Tables 1 and 2 and MIL-STD-1180A(AT) to determine which FMVSS are related to performance testing. Review previously selected tests to determine which performance criteria of the FMVSS have been or will be evaluated by other scheduled tests. Conduct the remaining tests for compliance in accordance with the appropriate FMVSS.

4.8.2 Data Required. Collect data as required by FMVSS and assess the impact of the observed performance on the overall safety of the vehicle.

4.9 Stowage.

4.9.1 Method. Use the procedures of TOP 2-2-802[25] to verify that stowage of materials is adequate, particularly with respect to vulnerability to fire and protection of personnel against fire.

4.9.2 Data Required. The identity, location and description of stowed items and stowage facilities are recorded. Assessments of the impact on safety are made considering flammability of materials, protection against accidental combustion, proximity to exhausts, interference with operational features, effects on stability, and retention of intended position.

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4.10 Safety Aspects of Maintenance (Required).

4.10.1 Method. Monitor safety aspects of vehicle maintenance functions throughout testing of the vehicle as required by TECOM Supplement 1 to DARCOM 700-15[26], watching for mechanical and electrical hazards (DARCOM Pamphlet 706-134[27]) and dangers that may arise from fire, hot temperatures, toxic gases, etc. Perform inspections such as the following:

a. Inspect the test item to verify that it is provided with necessary drains, pressure relief valves, electrical interlocks, cutoff switches, or other safety features and that such features are adequate and operable. Record non-operable safety features and the need for additional features.

b. Inspect warning plates and instruction plates for adequacy and conspicuous location. Record inadequate warnings and instructions, and recommend repositioning of plates if necessary. Ascertain whether the warning, caution, and instruction plates are applicable and conform to ANSI Z35.1[28].

c. Inspect the test item to determine whether suitable guards, covers, and enclosures are provided to shield personnel from accidental contact with dangerously exposed moving parts, electrical circuits, hazardous hot or cold surfaces, or injurious chemicals. Determine whether adequate personnel protective equipment (such as masks, safety goggles or face shields, hearing protectors, and gloves) is provided to protect the respiratory system, sight, hearing, and skin from toxic or noxious vapors or fumes, excessive noise, and thermal or chemical burns. Determine whether equipment publications contain cautions and instructions for the use of the required protective equipment. Record findings, classify hazard levels in accordance with MIL-STD-882A[29] and recommend any additional protection needed to safely maintain the item.

d. Inspect the provisions for lifting, jacking, towing, and access for freedom from safety hazards. Record findings and recommend improved safety features as necessary.

During the performance of maintenance, note the adequacy of safety provisions, including such features as stability of maintenance stands, adequacy of climbing and reaching provisions, placement of components, safeguards against reassembly in a manner that will result in subsequent unsafe operations, and sufficiency of precautions stated in the technical manuals.

4.10.2 Data Required. Data include observations noted above and narrative assessment indicating the adequacy or insufficiency of provisions or conditions to permit safe performance of maintenance on the vehicle. Report deficiencies and recommended remedial actions in detail.

4.11 Combat-and Tactical-Vehicle Considerations.

4.11.1 Method. Many of the basic safety evaluation procedures, including those in FMVSS, apply, particularly on new procurement or development vehicles for which MIL-STD-1180A(AT) directs compliance with FMVSS requirements.

Special considerations include, but are not limited to, the following:

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a. Crews must be adequately protected from the blast of the weapons they fire from the vehicle. Refer to MTP/TOP 2-2-625[30] for procedure to measure blast pressure in the crew compartment.

b. Hatch cover fasteners for both open and closed positions must lock positively and have adequate strength to withstand shock loads in either position. This can be determined using procedures of TOP/MTP 2-2-625.

c. Hazards arising from turret or gun movement will be minimized by preventive devices.

d. Brow pads must be sufficient to prevent injury or flinching (see TOP 1-2-610).

e. Whether the crew has means of egress, if vehicle overturns, can be ascertained by inspection in most instances.

f. The adequacy of hardware and procedures proposed to prevent injury to persons in and out of the vehicle will be evaluated by AEHA if LASER devices are to be used on the vehicle.

g. The mountings of all components attached to the inside surfaces of external walls will be carefully examined to assure the ballistic shock of non-penetrating projectiles will not cause the component to become a secondary missile within the crew compartment. Refer to TOP 2-2-620[31] for applicable test procedures.

h. Gun-firing systems will be examined and evaluated to assure that:

(1) Gun cannot be fired in an unsafe out-of-battery position.
(2) Firing can be prevented by a crewman who could be injured from weapon recoil.
(3) Circuits, equipment and procedures related to firing circuits of electrically fired weapons are reasonably safe from premature firing.

i. All ammunition racks will be examined to assure they prevent damage to the case and projectile.

j. The mounting of all electrical wiring near a moving part will be examined to assure it will not be abraded, given the quality of maintenance that can reasonably be expected during the period of time between depot rebuilds. Wiring under turret baskets will be given special consideration.

k. The adequacy of specified precautions to avoid injury from spent brass will be evaluated.

l. The level and nature of the potential hazard to crewmen from moving guns and turrets, especially in stabilized modes, will be described by the tester for consideration by user and developer.

m. Crew compartment must be adequately ventilated and crews must be adequately protected from toxic hazards arising from weapon firing. Refer to TOP 2-2-614 for test procedures for measuring toxic fumes.

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4.11.2 Data Required. Data such as the following are recorded: arming distance of projectile, protection from projectile blast and fragments, whether hatch covers lock positively, weapon mount stops, adequacy of brow pads, and means of egress.

4.12 Fire Extinguisher Systems.

4.12.1 Hand-Portable Extinguishers.

4.12.1.1 Method. This subtest is included to assure that hand-portable, dry-chemical, fire extinguishers, when required for a particular vehicle, meet the requirements of MIL-O-E-915[32]. Determine their accessibility and ease of operation within the context of the overall human factors evaluation. Weigh, discharge, and reweigh one extinguisher of each type.

4.12.1.2 Data Required. The number, size, and type of fire extinguishers and their net weight filler material will be recorded. Human factors difficulties are reported in HFE questionnaires.

4.12.2 Fixed Fire Extinguishers.

4.12.2.1 Method. Weigh and then install a first-shot extinguisher bottle for the crew compartment. With the engine idling, pull the internal control. Record the time when the control is activated, and the time that the discharge begins and ends. Remove the discharged extinguisher bottle and weigh it. Replace with another bottle before resuming testing. Check the electrical integrity of the fire-suppression system during the preoperational inspection and during regularly scheduled preventive maintenance. Evaluate the accessibility and ease of identifying and activating the manual control within the context of an overall human factors evaluation.

4.12.2.2 Data Required. Record the net weight of extinguisher that is discharged, and the duration of extinguishant discharge. Answers to HFE questionnaires pertinent to fire extinguishing system are required.

4.13 Safety Guards.

4.13.1 Methods. Provide safety guards for the protection of vehicle operator, occupants, or vehicle components. In some test operations, it may be desirable to install additional guards to minimize hazards to test participants. Examples are the overhead guards for cross-country operations over jungle trails to protect occupants of "soft-top" or "no-top" tactical vehicles from falling limbs, coconuts, etc., and brush guards for protection of windshield and lights. Overhead guards and rollover protective equipment apply particularly to forklifts, material-handling equipment and construction equipment. Conduct this test to assure that the guards provide adequate protection.

Use an applied static load, dropped test weights, or both, based on the required strength of the guard as prescribed in the specification or ROC. Procedures such as those described in ANSI B 56.1-1975[33] and SAE J231[34] are suitable for testing the performance of the overhead guard. 29 CFR 1910.1000 is suitable for testing rollover protective structures. Manufacturer's certification data may be accepted instead of testing the item.

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4.13.2 Data Required. Data to be collected include a description of guard and load, manner of application, and extent of damages. Data supplied by the manufacturer may be accepted in lieu of testing.

5. DATA REDUCTION AND PRESENTATION. Specific procedures for reducing and presenting data are explained in each of the TOP/MTPs cited in this TOP. As most subtests described in the referenced TOP/MTPs are not conducted to singularly evaluate safety and health aspects of vehicles, each subtest must be reviewed to recognize the safety and health implications of the test results. From this review, a concise listing of all existing and potential hazards can be compiled. The potential consequence of each hazard should be considered concurrently.

Each hazard is to be categorized with respect to severity and probability according to the provisions of MIL-STD-882A. When this is accomplished, each system characteristic that creates a hazard can more easily be classified as a deficiency or shortcoming following the guidance of TOP 1-1-012[35] and in accordance with DARCOM-R 70-13[36]. A worksheet to present this information is included in Appendix B. (A copy of this completed worksheet can become part of the detailed test plan and the final report.)

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APPENDIX A
CHECKLIST GUIDE FOR AUTOMOTIVE SAFETY EVALUATION

ITEM	YES	NO	NA	COMMENTS
VEHICLE				
1. Controls are accessible by the driver and easily identifiable.				
2. Transmission has a starter interlock and provides supplemental braking.				
3. Windshield wipers cover an adequate area and have two wiping speeds.				
4. Equipped with adequate defrosting/defogging systems				
5. Brakes hoses, glazing material, and tires are marked with manufacturer's certification of conformance to motor vehicle safety standards (symbol DOT, etc.).				
6. Metal components in the field of view of the driver have a non-reflective finish.				
7. Windshield and windows are configured to minimize glare.				
8. Lamps, reflectors, and associated signaling equipment conform to MIL-STD-1179[37].				
9. Rearview mirrors provide a clear and reasonably unobstructed view to the rear.				
10. Hoods are provided with a latch system. Front-opening hoods have a second-latch position or a second-latch system.				
11. Accelerator control has a backup, throttle-return spring.				
12. Vehicle is equipped with three, triangular, reflective, warning devices.				

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ITEM	YES	NO	NA	COMMENTS
13. Occupant crash protection (seatbelts, padding, etc.) is provided.				
14. Vehicle sideboards and tailgate are adequate to carry loose cargo and prevent cargo from slipping off the deck.				
15. Safety of personnel is not jeopardized by sharp, pointed, or hazardous projections.				
16. Maintenance access openings are free of burrs and sharp edges.				
17. Adjustment screws, controls, and replaceable parts are located away from high-voltage or high-temperature compartments.				
18. Non-slip material is provided in foot-traffic areas.				
19. Hatches have a positive retention device for the open position.				
20. Placards are provided for hazardous components or conditions.				
21. Hazardous components (electrical wiring, exhaust pipes, etc.) are located or protected to prevent use as a handhold.				
22. Fire extinguishers provided and of the correct type?				
23. Fire extinguishers readily accessible, but not adjacent to points where fires would probably originate?				
24. Critical warning lights isolated from other less important lights?				

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ITEM	YES	NO	NA	COMMENTS
25. Neutral parts of electrical systems grounded or isolated from contact with personnel or tools?				
26. Jacking and hoisting points clearly and conspicuously marked?				
27. Switches or controls which initiate hazardous operations require the prior operation of a related or locking control.				
28. Exhausts vented away from personnel inclosures.				
29. Multi-piece rim wheels are free from defects that might cause components to explode.[38]				

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ITEM	YES	NO	NA	COMMENTS
WINCHES/CRANES				
1. Cable size provides an adequate factor of safety.				
2. Adequate guards are provided to protect operator from cable failure and protect personnel from nip points.				
3. Hazardous moving parts are either fully enclosed or properly guarded.				
4. Snatch blocks, pulleys, slings, etc. are compatible with the designated cable size and load capacity.				
5. Cable is free of kinks and unstranded wires.				
6. Hook is equipped with a safety latch.				
7. An overload protection device is provided.				
8. A capacity chart indicating boom length, boom angle, and maximum load capacity is conspicuously posted.				
9. Suitable fire extinguisher in cab of crane				
10. Outriggers operate properly				
11. Adequate safe access to and from cab is provided.				
12. Warning is provided to prevent personnel from being exposed to swing of cab.				

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APPENDIX B
EQUIPMENT-OPERATION HAZARD ANALYSIS

Operating-hazard analysis may be performed during development testing to identify hazards associated with equipment, procedures, and personnel. Data from all development subtests can be used to prepare the Equipment Operation Hazard Analysis.

Figure B-1 is a typical Equipment Operation Hazard Analysis Worksheet. This worksheet is designed to assure a complete analysis and classification of hazards that have been identified. Whenever practical, it should become part of the DTP and/or the final report. The following instructions apply to the worksheet:

a. Hazard Description - Describe the personnel error, environmental condition, design inadequacy, procedural deficiency, system or component malfunction that presents a hazard to personnel, equipment, or property.

b. Hazard Effect - Describe the worst potential consequences to operating or maintenance personnel, equipment, or property, should the hazard continue to exist.

c. Hazard Category - Categorize the hazard in accordance with the provisions of MIL-STD-882A; this is accomplished in two parts. First, consider the hazard effect described in the second column of the worksheet. Based on this description, assign one of four possible hazard categories shown below:

(1) CATEGORY I - CATASTROPHIC: May cause death or system loss.

(2) CATEGORY II - CRITICAL: May cause severe injury, severe occupational illness, or major system damage.

(3) CATEGORY III - MARGINAL: May cause minor injury, minor occupational illness, or minor system damage.

(4) CATEGORY IV - NEGLIGIBLE: Will not result in injury, occupational illness, or system damage.

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HAZARD DESCRIPTION	HAZARD EFFECT	HAZARD CATEGORY/ PROBABILITY	HAZARD CLASSIFICATION	HAZARD CONTROLS AND REMARKS

Figure B-1. Typical equipment operation hazard analysis worksheet.

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After the hazard category is assigned, the qualitative probability that the hazard effect will occur in a specific individual item or in the Army inventory must be assigned. One of six possible hazard probability levels must be assigned from those listed below:

Hazard Probability

Descriptive Word	Level	Specific Individual Item	Inventory
Frequent	A	Likely to occur frequently	Continuously
Reasonably Probable	B	Will occur several times in life cycle of an item	Will occur frequently
Occasional	C	Likely to occur sometime in life cycle of an item	Will occur several times
Remote	D	So unlikely it can be assumed that this hazard will not be experienced	Unlikely to occur but possible
Extremely Improbable	E	Probability of occurrence cannot be distinguished from zero	So unlikely it can be assumed it will not be experienced
Impossible	F	Physically impossible to occur	Physically impossible to occur

Together, the hazard category (severity) and the hazard probability completely classify the hazard in accordance with MIL-STD-882A. For example, a critical hazard that occurs occasionally is a Category II-C hazard. This designation should be entered in the third column of the worksheet.

To classify the hazard as a deficiency, shortcoming, or suggested improvement, the Hazard Classification Guidelines provided as Figure B-2 may be used. This classification may be entered in the fourth column of the worksheet.

d. Hazard Controls and Remarks - Comments relative to what should be done to prevent the hazard or protect against the consequences should be included in the fifth column of the worksheet.

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		HAZARD PROBABILITY					
		FREQUENT	REASONABLY PROBABLE	OCCASIONAL	REMOTE	EXTREMELY IMPROBABLE	IMPOSSIBLE
SPECIFIC INDIVIDUAL ITEM		Likely to occur frequently	Will occur several times in life of item	Likely to occur sometime in the life of item	So unlikely, can be assumed that this hazard will not be experienced	Probability of occurrence cannot be distinguished from zero	Physically impossible to occur
FLEET OR INVENTORY		Continuously experienced	Will occur frequently	Will occur several times	Unlikely to occur, but possible	So unlikely, can be assumed that this hazard will not be experienced	Physically impossible to occur
		A	B	C	D	E	F
HAZARD SEVERITY: CATASTROPHIC - May cause death or system loss	I	DEFICIENCY	DEFICIENCY	DEFICIENCY	DEFICIENCY	SUGGESTED IMPROVEMENT OR ACCEPTABLE	ACCEPTABLE
HAZARD SEVERITY: CRITICAL - May cause severe injury or illness, or major system damage	II	DEFICIENCY	DEFICIENCY	DEFICIENCY	SHORTCOMING	SUGGESTED IMPROVEMENT OR ACCEPTABLE	ACCEPTABLE
HAZARD SEVERITY: MARGINAL - May cause minor injury or illness, or minor system damage	III	DEFICIENCY	SHORTCOMING	SHORTCOMING	SUGGESTED IMPROVEMENT	SUGGESTED IMPROVEMENT OR ACCEPTABLE	ACCEPTABLE
HAZARD SEVERITY: NEGLIGIBLE - Will not result in injury or illness, or system damage	IV	ACCEPTABLE	ACCEPTABLE	ACCEPTABLE	ACCEPTABLE	ACCEPTABLE	ACCEPTABLE

Figure B-2. Hazard probability versus hazard severity.

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APPENDIX C
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